



Chapter 3: Whole Numbers

Three-week revision booklet • 6 worksheets • 180 questions

NSW syllabus outcome

MA4-IND-C-01 — operates with primes, square roots and indices •

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How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Whole Numbers topic test. Each sheet has the same shape: **30 questions**, split into ● **Easy**, ● **Medium** and ● **Challenging**. Work from the top of the page down — the questions get harder as you go. There is a ruled line under the medium and challenging questions — use it to set out your working. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Rounding and estimating
1	2	Multiplying and dividing
2	3	Divisibility tests
2	4	Factors, multiples, HCF and LCM
3	5	Indices, roots and primes
3	6	Mixed revision — all skills

Progress tracker

Worksheet	Date completed	Score / 30	Checked by
1 Rounding and estimating			
2 Multiplying and dividing			
3 Divisibility tests			
4 Factors, multiples, HCF and LCM			
5 Indices, roots and primes			
6 Mixed revision — all skills			

Worksheet 1: Rounding and estimating

Week 1 • Worksheet 1 of 6 • 30 questions

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Look at the digit just to the right of the place you are rounding to. If it is 5 or more, round up. If it is 4 or less, round down.

EASY

15 QUESTIONS

Round each number to the nearest ten.

- | | | |
|--------------|-------------|--------------|
| 1. 47 _____ | 2. 82 _____ | 3. 135 _____ |
| 4. 268 _____ | 5. 91 _____ | 6. 574 _____ |

Round each number to the nearest hundred.

- | | | |
|---------------|----------------|---------------|
| 7. 372 _____ | 8. 850 _____ | 9. 1240 _____ |
| 10. 615 _____ | 11. 4380 _____ | 12. 927 _____ |

Round each number to the nearest thousand.

- | | | |
|----------------|----------------|------------------|
| 13. 6200 _____ | 14. 3480 _____ | 15. 12 750 _____ |
|----------------|----------------|------------------|

MEDIUM

9 QUESTIONS

Estimate each answer by rounding first.

- | | | |
|-----------------------------|-----------------------------|-----------------------------|
| 16. $38 + 51$
..... | 17. $297 + 104$
..... | 18. $412 - 198$
..... |
| 19. 62×9
..... | 20. 89×11
..... | 21. 51×19
..... |
| 22. 21×48
..... | 23. $179 + 322$
..... | 24. $608 - 291$
..... |

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|---|--|
| 25. Round 68 342 to the nearest thousand.
..... | 26. A stadium holds 48 617 people. Round this to the nearest thousand.
..... |
| 27. Estimate the cost of 19 items at \$5 each.
..... | 28. Estimate 6×397 . Is your estimate above or below the true value?
..... |
| 29. A number rounds to 500 to the nearest hundred. What is the smallest it could be?
..... | 30. Explain why estimating is useful before a long calculation.
..... |

Worksheet 2: Multiplying and dividing

Week 1 • Worksheet 2 of 6 • 30 questions

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Set your working out neatly in columns. Estimate first so you can check your answer is sensible.

EASY

15 QUESTIONS

Evaluate each product.

1. 34×6 _____

2. 23×4 _____

3. 41×5 _____

4. 62×3 _____

5. 18×9 _____

6. 27×2 _____

7. 45×6 _____

8. 36×5 _____

9. 52×7 _____

Evaluate each quotient.

10. $96 \div 8$ _____

11. $84 \div 7$ _____

12. $150 \div 5$ _____

13. $132 \div 4$ _____

14. $210 \div 6$ _____

15. $243 \div 9$ _____

MEDIUM

9 QUESTIONS

Evaluate each expression.

16. 246×9

17. 128×4

18. 305×6

19. $936 \div 6$

20. $455 \div 5$

21. $816 \div 4$

22. 72×20

23. $1200 \div 8$

24. 45×12

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

25. A school buys 24 boxes of 30 pencils. How many pencils is that?

26. \$480 is shared equally between 12 people. How much does each get?

27. A truck carries 18 pallets of 145 bricks. How many bricks altogether?

28. Tickets cost \$25. How many can be bought with \$400?

29. A rectangle is 26 cm by 15 cm. Find its area.

30. A number multiplied by 15 gives 300. Find the number.

Worksheet 3: Divisibility tests

Week 2 • Worksheet 3 of 6 • 30 questions

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A number is divisible by **2** if it is even, by **5** if it ends in 0 or 5, by **10** if it ends in 0, by **3** if its digits add to a multiple of 3, and by **9** if its digits add to a multiple of 9.

EASY

15 QUESTIONS

Is each number divisible by 2? Write yes or no.

- | | | |
|--------------|---------------|--------------|
| 1. 146 _____ | 2. 273 _____ | 3. 580 _____ |
| 4. 999 _____ | 5. 1024 _____ | 6. 365 _____ |

Is each number divisible by 5? Write yes or no.

- | | | |
|---------------|---------------|----------------|
| 7. 145 _____ | 8. 230 _____ | 9. 612 _____ |
| 10. 875 _____ | 11. 404 _____ | 12. 1000 _____ |

Is each number divisible by 10? Write yes or no.

- | | | |
|---------------|---------------|----------------|
| 13. 470 _____ | 14. 365 _____ | 15. 1200 _____ |
|---------------|---------------|----------------|

MEDIUM

9 QUESTIONS

Test each number for divisibility by the number in brackets.

- | | | |
|----------------------|----------------------|----------------------|
| 16. 216 (3)
_____ | 17. 534 (3)
_____ | 18. 721 (3)
_____ |
| 19. 342 (9)
_____ | 20. 819 (9)
_____ | 21. 455 (9)
_____ |
| 22. 128 (4)
_____ | 23. 316 (4)
_____ | 24. 530 (4)
_____ |

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|---|--|
| 25. List all the numbers from 2 to 10 that 360 is divisible by.

_____ | 26. Find the smallest number greater than 200 that is divisible by 9.

_____ |
| 27. Find the largest three-digit number divisible by 4.

_____ | 28. A number is divisible by both 2 and 3. What else must it be divisible by?

_____ |
| 29. The number 4_2 is divisible by 3. Find every possible missing digit.

_____ | 30. Explain why every number divisible by 9 is also divisible by 3.

_____ |

Worksheet 4: Factors, multiples, HCF and LCM

Week 2 • Worksheet 4 of 6 • 30 questions

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The **HCF** is the largest factor two numbers share. The **LCM** is the smallest multiple they share.

EASY

15 QUESTIONS

List all the factors of each number.

- | | | |
|-------|-------|-------|
| 1. 12 | 2. 18 | 3. 20 |
| _____ | _____ | _____ |
| 4. 24 | 5. 15 | 6. 30 |
| _____ | _____ | _____ |

Write the first four multiples of each number.

- | | | |
|-------|-------|-------|
| 7. 3 | 8. 5 | 9. 7 |
| _____ | _____ | _____ |
| 10. 4 | 11. 6 | 12. 9 |
| _____ | _____ | _____ |

Is each number prime or composite?

- | | | |
|--------------|--------------|--------------|
| 13. 11 _____ | 14. 21 _____ | 15. 29 _____ |
|--------------|--------------|--------------|

MEDIUM

9 QUESTIONS

Find the highest common factor (HCF).

- | | | |
|---------------------|---------------------|---------------------|
| 16. 8 and 12 _____ | 17. 15 and 25 _____ | 18. 18 and 30 _____ |
| 19. 12 and 20 _____ | 20. 9 and 27 _____ | 21. 14 and 35 _____ |

Find the lowest common multiple (LCM).

- | | | |
|-------------------|-------------------|-------------------|
| 22. 4 and 5 _____ | 23. 6 and 9 _____ | 24. 3 and 7 _____ |
|-------------------|-------------------|-------------------|

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 25. Find the HCF of 24 and 36.
_____ | 26. Find the LCM of 8 and 12.
_____ |
| 27. Write 60 as a product of prime factors.
_____ | 28. Two bells ring every 6 and 8 minutes. How often do they ring together?
_____ |
| 29. List all the factors of 48.
_____ | 30. Explain why the HCF of two prime numbers is always 1.
_____ |

Worksheet 5: Indices, roots and primes

Week 3 • Worksheet 5 of 6 • 30 questions

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3^2 means 3×3 . $\sqrt{25}$ asks: which number multiplied by itself gives 25?

EASY

15 QUESTIONS

Evaluate each expression.

1. 2^3 _____

2. 5^2 _____

3. 3^2 _____

4. 4^2 _____

5. 10^2 _____

6. 6^2 _____

7. 2^4 _____

8. 1^5 _____

9. 7^2 _____

Evaluate each square root.

10. $\sqrt{9}$ _____

11. $\sqrt{16}$ _____

12. $\sqrt{25}$ _____

13. $\sqrt{36}$ _____

14. $\sqrt{49}$ _____

15. $\sqrt{64}$ _____

MEDIUM

9 QUESTIONS

Evaluate each expression.

16. $\sqrt{81}$ _____

17. $\sqrt{100}$ _____

18. $\sqrt{121}$ _____

19. $\sqrt{144}$ _____

20. 2^5 _____

21. 3^3 _____

22. 10^3 _____

23. Write $4 \times 4 \times 4$ in index form _____

24. Write 5×5 in index form _____

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

25. Evaluate $3^2 + 4^2$.

26. Evaluate $\sqrt{36} + \sqrt{64}$.

27. List the prime numbers between 20 and 40.

28. Write 47 as a Roman numeral.

29. Which is larger, 2^4 or 3^3 ?

30. Between which two whole numbers does $\sqrt{50}$ lie?

Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • 30 questions

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This sheet mixes every skill from the Whole Numbers chapter. Work through it as though it were the topic test.

EASY

15 QUESTIONS

Evaluate each expression.

- | | | |
|------------------------|------------------------|------------------------|
| 1. 34×6 _____ | 2. $96 \div 8$ _____ | 3. 5^2 _____ |
| 4. $\sqrt{49}$ _____ | 5. 45×5 _____ | 6. $132 \div 4$ _____ |
| 7. 2^3 _____ | 8. $\sqrt{100}$ _____ | 9. 27×3 _____ |

Round 57 486 to the nearest:

- | | | |
|---------------|-------------------|--------------------|
| 10. ten _____ | 11. hundred _____ | 12. thousand _____ |
|---------------|-------------------|--------------------|

Is each number divisible by 3? Write yes or no.

- | | | |
|---------------|---------------|---------------|
| 13. 216 _____ | 14. 451 _____ | 15. 630 _____ |
|---------------|---------------|---------------|

MEDIUM

9 QUESTIONS

Answer each question.

- | | | |
|-------------------------------------|-----------------------------------|--|
| 16. HCF of 12 and 18
..... | 17. LCM of 4 and 6
..... | 18. List the factors of 20
..... |
| 19. Estimate 62×9
..... | 20. Estimate $297 + 104$
..... | 21. Is 480 divisible by 4?
..... |
| 22. Evaluate $3^2 + 4^2$
..... | 23. Evaluate $\sqrt{81}$
..... | 24. Write $3 \times 3 \times 3$ in index form
..... |

CHALLENGING

6 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 25. List the prime numbers between 20 and 40.
..... | 26. Write 60 as a product of prime factors.
..... |
| 27. Find the HCF and the LCM of 8 and 12.
..... | 28. A box holds 24 cans. How many boxes are needed for 300 cans?
..... |
| 29. Write 47 as a Roman numeral.
..... | 30. The number 5_1 is divisible by 3. Find every possible missing digit.
..... |